INCH-POUND

MIL-H-28692(YD)
6 April 1993

MILITARY SPECIFICATION

# HOSE, NONMETALLIC, POTABLE WATER DISCHARGE, SMOOTH-BORE, LIGHTWEIGHT, 4 INCH, WITH REATTACHABLE COUPLINGS

This specification is approved for use by the Naval Facilities Engineering Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

## 1. SCOPE

1.1 Scope. This specification covers a 4-inch, lightweight, smooth-bore, potable water discharge hose, suitable for use within a temperature range of -25 to 130 degrees Fahrenheit ($^{o}$F), furnished with couplings and a hose support ring. The hoses and couplings are intended for use as components of a 10,000-foot ship-to-shore buoyant potable water supply system. Storage of the hoses, while not in use, will be in a collapsed state on ship mounted hosereels exposed to weather.

## 2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. The following specifications and standards form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commanding Officer (Code 156), Naval Construction Battalion Center, 621 Pleasant Valley Road, Port Hueneme, CA 93043-4300, by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 4720

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SPECIFICATIONS

MILITARY

MIL-H-775 - Hose, Hose Assemblies: Rubber, Plastic, Fabric or Metal (Including Tubing) and Associated Hardware: Packaging of.
MIL-C-24356 - Coupling, Segmented and Split Clamps, Reattachable, 2-1/2, 4-, 6-, and 7-inch, for Refueling-at-Sea Hose.

STANDARDS

FEDERAL

FED-STD-162 - Hose, Rubber, Visual Inspection Guide for.
FED-STD-601 - Rubber; Sampling and Testing.

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.

(Copies of specifications, standards, publications, and other Government documents required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

2.2 Non-Government publications. The following document(s) form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted are those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issues of the documents which are current on the date of the solicitation (see 6.2).

AMERICAN PUBLIC HEALTH ASSOCIATION, INC. (APHA)

Standard Methods for the Examination of Water and Wastewater.

(Application for copies should be addressed to the American Public Health Association, Inc., 1015 5th Street N.W., Washington, DC 20005).

ASTM

ASTM A153 - Zinc Coating (Hot-Dip) on Iron and Steel Hardware.
ASTM A572 - High-Strength Low-Alloy Columbium-Vanadium Steels of Structural Quality.
ASTM D380 - Testing Rubber Hose.
ASTM D412 - Rubber Properties in Tension.
ASTM D413 - Rubber Property - Adhesion to Flexible Substrate.
ASTM D471 - Rubber Property - Effect of Liquids.
ASTM D518 - Rubber Deterioration - Surface Cracking.
ASTM D1149 - Rubber Deterioration - Surface Ozone Cracking in a Chamber.
ASTM D1193 - Reagent Water.



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(Application for copies should be addressed to the ASTM, 1916 Race Street, Philadelphia, PA 19103.)

(Non-Government standards and other publications are normally available from the organizations which prepare or which distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein (except for associated detail specifications, specification sheets or MS standards), the text of this specification shall take precedence. Nothing in this specification however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified in the contract or purchase order, a complete hose assembly shall be subjected to first article inspection (see 4.2.1 and 6.2).

3.2 Components. The complete hose assembly shall consist of the basic hose and two reattachable couplings (a male coupling attached to one end of the hose, and a female coupling attached to the other end).

3.3 Materials. Materials shall be as specified herein and in applicable specifications and standards, and other referenced documents. Materials not specified shall be selected by the contractor and shall be subject to all provisions of this specification. Materials shall be free of defects which adversely affect performance or serviceability of the finished product.

3.3.1 Tube. The inner tube shall be compounded for use with potable water. The tube material shall be approved by the Food and Drug Administration (FDA), the Environmental Protection Agency (EPA), or the National Sanitation Foundation (NSF) for contact with potable water. The tube shall be seamless.

3.3.2 Cover. The cover shall be synthetic rubber or thermoplastic urethane (TPU) compounded for high resistance to sea water, fresh water, ozone attack, oil, weathering in the temperature range stated, and abrasion. Synthetic rubber selected for the cover material shall have performance properties equal to or greater than polychloroprene for the conditions stated herein.

3.3.3 Excluded material. The tube and cover shall not contain ester based polyurethane. The material shall not display the strongest infrared spectral absorption band in the wave number, reciprocal centimeters ($cm^{-1}$), region 1700 - 1750 $cm^{-1}$ when tested in accordance with 4.5.16.

3.4 Construction. The hose construction shall consist of TPU extruded through a woven reinforcement or alternately, a bonded synthetic rubber construction using two or more plies of reinforcement. All hoses shall have a minimum service life of five years, 90 percent of the time stored on the ship's hosereel and covered with a tarpaulin, and 10 percent in-service.



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3.4.1 Extruded TPU construction. The hose shall be constructed of a woven synthetic fiber reinforcement completely encased in an extruded TPU so that an elastomeric bond through the interstices of the weave will provide a continuous elastomeric bond from the inner tube through to the outer cover.

3.4.2 Bonded synthetic rubber construction. The hose shall have a rubber tube, reinforcement consisting of either braided synthetic fiber yarn or spiral plies of synthetic fiber cord held together with rubber, friction impregnated through the cord so as to create a rubber-to-rubber bond, and a rubber cover. All layers shall be bonded to their adjacent layers so as to produce a unified hose wall. The hose shall have a longitudinal reinforcement of synthetic fiber cord incorporated into the hose carcass and extending the length of the hose. Both ends of the hose shall be cut square.

3.5 Dimensions. The hose shall have an inside diameter (id) of 4 inches +0.062 -0.030 inch. Each hose section shall have a continuous length of 150 ±1.5 feet. If the hose is constructed using a synthetic rubber compound conforming to the requirements of this specification, then the cover thickness of the hose shall be 0.035 inch minimum when measured in accordance with 4.5.1. If the hose cover is constructed with a TPU compound, the cover thickness of the hose shall be 0.020 inch minimum when measured in accordance with 4.5.1.

3.6 Weight. The unit weight of the hose, without couplings, shall not exceed 2.0 pounds (lb) per linear foot when measured in accordance with 4.5.2.

3.7 Strength.

3.7.1 Hydrostatic pressure. The design working pressure of the hose shall be 150 pound-force per square inch gage (psig). The hose shall withstand a proof pressure of 300 psig without leakage or other indication of weakness when tested in accordance with 4.5.3. The hose shall not burst at a pressure less than 600 psig when tested in accordance with 4.5.5.

3.7.2 Dimensional stability. When tested in accordance with 4.5.4, the hose shall not increase in length more than 4 percent nor increase in outside diameter (od) more than 12 percent, nor shall the twist be more than $1^{o}$ per foot while under the proof pressure. The hose shall not retain more than 2 percent change in length or 2 percent change in od after releasing the pressure. All calculations shall be based on the original dimensions of the hose.

3.7.3 Tensile pull and elongation. The hose shall be capable of withstanding a tensile pull of 8,000 lb for 5 minutes and shall have a breaking tensile of not less than 10,700 lbs when tested in accordance with 4.5.6. At the 8,000 lb tensile pull there shall be no apparent damage to the hose or weakening of the coupling grips. The hose shall have an elongation of not more than 10 percent when loaded to 8,000 lb in accordance with 4.5.6.

3.8 Collapsibility. When tested in accordance with 4.5.8, the hose shall collapse so that the lengths of the major and minor axis (figure 1) are in the "Region of Acceptability" in figure 2.

3.9 Flexibility. The flexibility of the hose shall be such that, when empty, it shall show no permanent deformation or leakage when tested in accordance with 4.5.9.



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3.10 Environmental exposure. The hose shall not crack, break, leak, or show permanent deformation when tested in accordance with 4.5.10.

3.11 Color. The color of the hose shall be blue.

3.11.1 Alternate color. An alternate hose color shall be black with blue tracer stripes that encircle or spiral the length of the hose, or four longitudinal stripes, approximately 90° apart, shall run the full length of the hose. The stripes shall be at least 0.25 inch wide. The circular rings or spirals shall be visible from a distance of 6 feet.

3.12 Taste and odor. The hose shall not impart taste to a rating level higher than 4 or impart odor to an intensity greater than threshold odor No. 2 to chlorinated (drinking) water when tested in accordance with 4.5.7.

3.13 Physical properties.

3.13.1 Adhesion.

3.13.1.1 Original adhesion. The original adhesion, between tube and plies, between the plies, and between the cover and the plies, shall not be less than 15 lb when measured in accordance with 4.5.12.

3.13.1.2 Adhesion after immersion in water. After immersion in water, the adhesion between the tube and plies, between the plies, and between the cover and the plies shall not be less than 10 lb when measured in accordance with 4.5.13.

3.13.2 Tensile strength. The original tensile strength of the tube shall not be less than 2,300 pound-force per square inch (psi) and the tensile strength of the cover shall not be less than 1,600 psi when measured in accordance with 4.5.11. The tensile strength of the tube and cover (after immersion in water), shall not be less than 65 percent of the tensile strength obtained before immersion.

3.13.3 Ultimate elongation. The original ultimate elongation of the tube and cover shall not be less than 300 percent, and the elongation after immersion in water shall not be less than 200 percent when measured in accordance with 4.5.11.

3.13.4 Ozone resistance. The cover shall show no visible cracking under 7X magnification after testing in accordance with 4.5.11.

3.13.5 Oil resistance. The volume increase of the cover after immersion in ASTM oil No. 3 shall not exceed 80 percent when measured in accordance with 4.5.11.

3.13.6 Extraction of inner tube. The extractable nonvolatile matter in inner tube shall not exceed 21 milligrams per square inch when tested as specified in 4.5.14.

3.14 Marking. Each length of hose shall be marked in color that contrasts with the color of the hose cover. The marking shall be accomplished either by inlaying a rubber or suitable material brand or by applying a suitable



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composition ink, bonding the marking onto the cover so that the marking cannot be removed except by mechanical means. The marking shall consist of the manufacturer's name or trademark, the quarter and year of manufacture, and the words "Potable Water Use Only." Hose shall be marked at intervals of at least 25 feet with letters at least 0.25 inch high.

3.14.1 Marking, alternate method. An alternate method of marking may be the application of a continuous embossed strip along the entire length, vulcanizing the hose with subsequent removal of the strip, leaving a continuous relief identification area. Identification shall be as specified in 3.14 and repeated at maximum intervals of 36 inches. When the marking is accomplished by this alternate method, no color contrast is required.

3.15 Couplings and fittings. Each of the hoses shall be equipped with couplings conforming to MIL-C-24356, size 4-inch, with the following exceptions: (1) Dimension J in figure 1 shall be no less than 3.400 inches, dimensions L and M in figure 1 shall be 1.53 inches and 1.59 inches to assure clearance for installation of hose support ring; (2) Material composition for end coupling body and nipple in table III shall be wrought aluminum alloy (6061-T6); and (3) Couplings are not required to be qualified under MIL-C-24356, but shall be provided by a manufacturer that has products qualified under MIL-C-24356. One coupling on each hose shall be the male type, the other coupling on each hose shall be the female type. The couplings, both male and female, shall be so designed that a hose support ring fits between the nipple and the back of the coupling in such a manner so that movement along the axis of the coupling is limited and rotation due to an applied tangential load is unrestricted when the coupling is engaged.

3.16 Split clamp and band assembly. One 4-inch split clamp and band assembly conforming to MIL-C-24356, consisting of "O" ring, two split clamp halves, one band and all necessary screws shall be provided with each length of hose.

3.17 Hose support ring (RHS-4). One hose support ring (RHS-4), conforming to figure 3, shall be provided per 150 foot section of hose for installation as shown therein. The inner diameter of the hose support ring shall not change greater than 5 percent with a load of 6,000 lb when tested in accordance with 4.5.15.

3.18 Workmanship.

3.18.1 Hose. The exterior surface of the hose shall have no more total visual defects that are allowed by 4.3.1 when examined in accordance with 4.4.

3.18.2 Interchangeability. All parts and fittings shall be fabricated and assembled so that units of the same type are interchangeable. Critical dimensions and tolerances for couplings and fittings shall be as shown on the applicable drawings and shall be strictly observed.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise



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specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items must meet all requirements of section 3. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.1.2 Component and material inspection. Components and materials shall be inspected in accordance with all the requirements specified herein and in applicable referenced documents.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.2.1).
b. Quality conformance inspection (see 4.2.2).

4.2.1 First article inspection. The first article inspection shall be performed on a complete hose assembly. This inspection shall include the examination of 4.4 and the tests of 4.5. The first article may be either a first production item or a standard production item from the supplier's current inventory provided the item meets the requirements of the specification and is representative of the design, construction, and manufacturing technique applicable to the remaining items to be furnished under the contract. In addition to the first article the following shall be submitted: A 36-inch length of hose for the burst test; a 12-inch length of hose for the collapsibility test; a 12-inch length of hose for the tensile pull and elongation test; an 8-foot length for the environmental test; and a 12-inch length for the taste and odor test.

4.2.2 Quality conformance inspection. The quality conformance inspection shall include the examination of 4.4, and the tests of 4.5 except 4.5.7 and 4.5.14 and the packaging inspection of 4.6. This inspection shall be performed on the samples selected in accordance with 4.3.

4.3 Sampling. Sampling and inspection procedures shall be in accordance with MIL-STD-105. The unit of product shall be one complete hose assembly. All hose assemblies offered for delivery at one time shall be considered a lot for the purpose of inspection.

4.3.1 Sampling for examination. Guidance for sampling procedures is provided in 6.5.1.



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4.3.2 Sampling for tests. Guidance for sampling procedures is provided in 6.5.2.

4.3.2.1 Sample for collapsibility test. A 12-inch length of hose identical to the hose submitted for the lot shall be submitted for the collapsibility test specified in 4.5.8.

4.3.2.2 Sample for destructive tests. A 36-inch length of hose identical to the hose submitted for the lot shall be submitted for the burst test specified in 4.5.5. At the completion of the burst test, additional tests shall be conducted as specified in 4.5.1.3, 4.5.11, 4.5.12, 4.5.13, and 4.5.14 on an undamaged section of the 36-inch sample of the hose submitted for the burst test.

4.3.2.3 Sample for tensile pull and elongation test. A 12-inch length of hose identical to the hose submitted for the lot shall be submitted for the tensile pull and elongation test specified in 4.5.6.

4.3.2.4 Sample for environmental test. An 8-foot length of hose identical to the hose submitted for the lot shall be submitted for the environmental test specified in 4.5.10.

4.3.2.5 Sampling for proof pressure test. Every hose in the lot shall be subjected to the nondestructive proof pressure test (see 4.5.3).

4.3.2.6 Sample for taste and odor test. A 12-inch length of hose identical to the hose submitted for the lot shall be submitted for the taste and odor test specified in 4.5.7.

4.4 Examination. Each sample hose assembly selected shall be visually and dimensionally examined to determine conformance with all the requirements of this specification not involving tests. The classification of defects in FED-STD-162 shall be used to determine and evaluate defects through visual inspection.

4.5 Tests. Sample hose sections selected shall be subjected to the applicable test specified in 4.5.1 through 4.5.14.

4.5.1 Dimensions.

4.5.1.1 Length. Hose selected shall be measured for length as specified in ASTM D380, to determine conformance to 3.5.

4.5.1.2 Inside diameter. Hose selected shall have the id measured as specified in ASTM D380 to determine conformance to 3.5.

4.5.1.3 Cover thickness. The thickness of the cover shall be measured on specimens removed from each end of the hose section obtained in accordance with 4.3.2.2 to determine conformance to 3.5. The thickness shall be determined by the procedures given in ASTM D380, except that only buffed specimens shall be measured. The respective values found for the cover thickness shall not be averaged, but shall be regarded as independent measurements, each of which should meet the requirement.



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4.5.2 Unit weight. The unit weight of all hoses in the sampling shall be determined by weighing the hoses without couplings. The readings shall be accurate to the closest pound. The weight of each hose shall be divided by its length to determine conformance to 3.6.

4.5.3 Proof pressure. Each length of hose shall be equipped with couplings and subjected to 300 psig proof pressure to determine conformance to 3.7.1. Before starting the tests, a 10 lb hydrostatic pressure shall be established in the hose, and the length, od, and zero index for the twist shall be measured (ASTM D380) in preparation for the dimensional stability test (see 4.5.4). The method shall be as described in ASTM D380. The proof pressure shall be held for 5 minutes and the hose and fittings examined for leakage or other indication of weakness. At the end of the 5-minute period and while the pressure is maintained, the length, od, and twist of the hose shall be measured for the determination of dimensional stability.

4.5.4 Dimensional stability. The effect of the proof test pressure on the length, od, and twist of all hoses in the sampling shall be determined as specified in 4.5.3 and ASTM D380. The pressure shall then be reduced to 0 psig and hose left in this condition for 5 minutes. At the end of this period, the diameter and length of the hose shall be measured to determine conformance to 3.7.2.

4.5.5 Burst pressure. A 36-inch section of the hose obtained in accordance with 4.3.2.2, shall be equipped with couplings and the bursting strength of this section in the straight condition shall then be determined in accordance with ASTM D380 to determine conformance to 3.7.1.

4.5.6 Tensile pull and elongation test. A section, measuring 12 inches in length between couplings, shall be obtained in accordance with 4.3.2.3 and equipped with couplings conforming to this specification on each end. The ends of the couplings on the hose shall be adapted for testing in tension in a universal testing machine. Two bench marks, 1 inch apart and 5-1/2 inches from each end of the hose between couplings, shall be marked on the outside (cover) for measuring elongation. The section of hose with couplings shall be subjected to a tensile load of 8,000 lb in the axial direction and held at that load while the increase in distance between the bench marks is measured. This measurement and the original measurement shall be used to calculate the elongation percent for conformance to 3.7.3. The tensile loading shall be held at 8,000 lb for 5 minutes while the hose is inspected for damage or weakening of the coupling grips. The tensile loading shall be released and the hose allowed a period of 10 minutes to return to normal. Visible damage, such as wrinkling or bubbling of the cover or interior tube, or a drastic increase in length or decrease in diameter indicating permanent set or delamination of the hose components, shall be cause for rejection. The tensile load shall then be reapplied and increased until failure occurs. The failure load and the type of failure, hose break, or coupling pull-out, shall be recorded. The rate of loading for the elongation and tensile tests shall be 1.0 inch per minute.

4.5.7 Taste and odor. Taste and odor test shall be performed in accordance with procedures outlined in American Public Health Association, Standard Methods for Examination of Water and Wastewater. A sample of the hose shall be filled with distilled water having 0.2 parts per million of total available chlorine at the start of the 70-hour test. The sample shall be long enough to expose 36



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square centimeters of inner tube to one liter of chlorinated distilled water. Nonconformance to 3.12 shall constitute failure of this test.

4.5.8 Collapsibility. The specimen for this test shall be a 12-inch section with square-cut ends. The load shall be applied perpendicular to the longitudinal direction of the specimen by means of two plates which evenly distribute the applied load over the entire specimen. The hose specimen shall be compressed at a rate of 1-inch per minute until the applied load has reached 67 ±1 lb force. Within 15 seconds after reaching this load, measure the lengths of the major and minor axis (see figure 1). Major and minor axis refer to hose outside dimensions. Failure of the lengths of the major and minor axis to lie in the "Region of Acceptability" of figure 2 shall be cause for rejection.

4.5.9 Flexibility test. The test shall be carried out on an empty length of hose equipped with couplings. The hose shall be coiled around a test drum having an external diameter of 36 inches. After uncoiling, the hose shall be proof tested as in 4.5.3.

4.5.10 Environmental test. A sample 8-foot length of hose shall be folded 180° and tied in the folded position. The sample shall be immersed in water conforming to type IV Reagent Water in accordance with ASTM D1193 at 158 ±3.6°F for a period of 14 days. The hose shall then be conditioned in a cold chamber for 72 hours at -25 ±3.6°F. The hose shall be removed from the test chamber and unfolded 180° using a force of not more than 10 lb within 5 seconds. The unfolding shall be accomplished within 3 minutes after removal from the test chamber. The hose sample shall then be hydrostatic pressure tested at 300 psig in accordance with 4.5.3.

4.5.11 Properties of rubber and TPU components. To determine conformance to 3.13, the properties of rubber components shall be determined on longitudinal specimens prepared from the hose section obtained in accordance with 4.3.2. After separating and buffing smooth, the tube and cover samples shall be permitted to rest not less than 30 minutes before preparing test specimens. Rubber samples shall be tested as follows:

| Property | Test Method |
|---|---|
| Initial ultimate elongation | ASTM D412 |
| Initial tensile strength | ASTM D412 |
| Ozone resistance | ASTM D1149 1/ |
| Tensile strength after immersion | ASTM D471 2/ |
| Ultimate elongation with immersion | ASTM D471 2/ |
| Volume increase after immersion | ASTM D471 3/ |

1/Specimens mounted in accordance with Method B, ASTM D518, ozone concentration shall be 100 ±5 parts per hundred million; time of exposure shall be 168 hours 104 ±3.6°F. Remove sample from oven at 80 hours, flex the sample twice by pushing the loop at zones 4 and 5 downward toward the base, reinstall it in the oven per Method B, and run test to completion.
2/The samples shall be immersed in distilled water for 14 days at 158 ±3.6°F.
3/Hose cover samples shall be immersed in ASTM Oil No. 3 for 70 hours at 212 ±3.6°F.



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Nonconformance to 3.13.2, 3.13.3, 3.13.4, or 3.13.5 shall constitute failure of these tests.

4.5.12 Original adhesion. From test section of hose selected under 4.3.2.2, prepare ring specimens as described in ASTM D413. The adhesion on the ring specimen shall be determined in accordance with ASTM D413, Machine method. Nonconformance to 3.13.1.1 shall constitute failure of this test.

4.5.13 Adhesion after immersion in water. Specimens prepared as described in 4.5.12 shall be immersed in distilled water for 14 days at 158 $\pm$3.6°F. Within 5 minutes after removal from immersion, the adhesion shall be determined as described in 4.5.12. Nonconformance to 3.13.1.2 shall constitute failure of this test.

4.5.14 Extraction of hose. The inner tube surface of one hose sample selected in accordance with 4.3.2.2 shall be extracted in distilled water at reflux temperature in accordance with the following procedures:

a. Remove a 1-inch ring of rubber inner tube and cut or buff away the outer surface until the surface is smooth.
b. Cut sufficient length of inner tube to obtain approximately 10 grams, based on a minimum thickness of 3/64 inch.
c. Subject sample to distilled water, at reflux temperature, for 7 hours in an extraction apparatus as shown in method 16001 of FED-STD-601.
d. Filter the solution through No. 40 filter paper and collect the extract in a tared container.
e. Extract the rubber sample in an additional 50-75 cubic centimeters of distilled water at a reflux temperature for 2 more hours and repeat step (d). Combine the extract solutions and evaporate to dryness. The container and residue shall be dried in an oven at 221 $\pm$3.6°F for one hour, cooled in a desiccator and weighed. The weight of the residue shall be recorded. A blank shall be run using the same amount of distilled water. After making allowance for the blank, the weight of the residue shall be recorded to the nearest milligram.
f. Report the amount of matter extracted from the rubber in milligrams per square inch of surface, determined as follows:

   Length (inches) x width (inches) x 2 = total surface area (sq. in.)
   Milligrams per square inch = weight of extract/total surface area.

4.5.15 Hose support ring test. Approval for the hose support ring for the 4-inch hose couplings will be based on the following test:

a. Using a Universal Tensile Testing Machine, a tensile testing harness will be prepared so that the bearing surface for the ring is circular having a maximum diameter of 4.33 inches. The bearing surface shall be held by the testing machine so that when loaded, it will not rotate or skew to one side. A swivel hook with a latch (with a working load rating of 4 tons) will be attached to one of the line attachment points on the ring. The swivel will be attached to a 2-foot long 5/8-inch wire rope sling using a shackle (with a working load rating of 4 tons). The other end of the sling should be attached to the testing machine in a safe and appropriate manner.


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b. Gradually load the ring to 6,000 lb. Gradually release the load. Repeat this operation nine additional times.
c. Load the ring to 6,000 lb and hold the load for 15 minutes. Release the load and examine the ring.

4.5.16 Fourier transform infrared (FT-IR) analysis. Samples of the tube and cover material shall be analyzed by the FT-IR method. Nonconformance to 3.3.3 shall constitute failure of this test.

4.6 Packaging inspection. The preservation, packing, and marking of the hose shall be inspected to the requirements of Section 4 of MIL-H-775.

5. PACKAGING

5.1 Preservation, packing, and marking. The hose shall be preserved, packaged, packed and marked in accordance with MIL-H-775 with the level of preservation and level of packing as specified (see 6.2).

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The hoses and couplings covered by this specification are intended for use as components of a 10,000-foot ship-to-shore buoyant potable water supply system. Storage of the hoses, while not in use, will be in a collapsed state on ship mounted hosereels exposed to weather.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.
b. Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1.1 and 2.2).
c. When first article is required for inspection and approval (see 3.1).
d. Level of preservation and level of packing required (see 5.1).

6.3 Data requirements. When this specification is used in an acquisition and data are required to be delivered, the data requirements shall be developed as specified by an approved Data Item Description (DD Form 1664) and delivered in accordance with the approved Contract Data Requirements List (CDRL), incorporated into the contract. When the provisions of DoD Federal Acquisition Regulations (FAR) Supplement, Part 27, Sub-Part 27.475-1 (DD Form 1423) are invoked and the DD Form 1423 is not used, the data should be delivered by the contractor in accordance with the contract or purchase order requirements.

6.4 First article. When a first article inspection is required, the item will be tested and should be a first production item or it may be a standard production item from the contractor's current inventory as specified in 4.2.1. The first article should consist of one hose assembly. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examination, test, and approval of the first article.


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6.5 Sampling procedures.

6.5.1 Sampling for examination. Recommended inspection level is II and Acceptable Quality Level is 1.0 for major defects and 4.0 for minor defects (see 4.3.1).

6.5.2 Sampling for tests. Recommended inspection level is S-3 and Acceptable Quality Level is 4.0 (see 4.3.2).

6.6 Subject term (key word) listing.

Buoyant
Hosereels
Hydrostatic tests
Support ring

Custodian:
Navy - YD

Preparing Activity:
Navy - YD

(Project 4720-N048)



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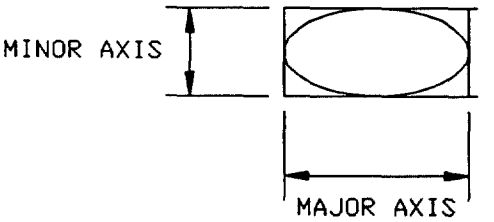



FIGURE 1. Major and minor axis of collapsed hose.

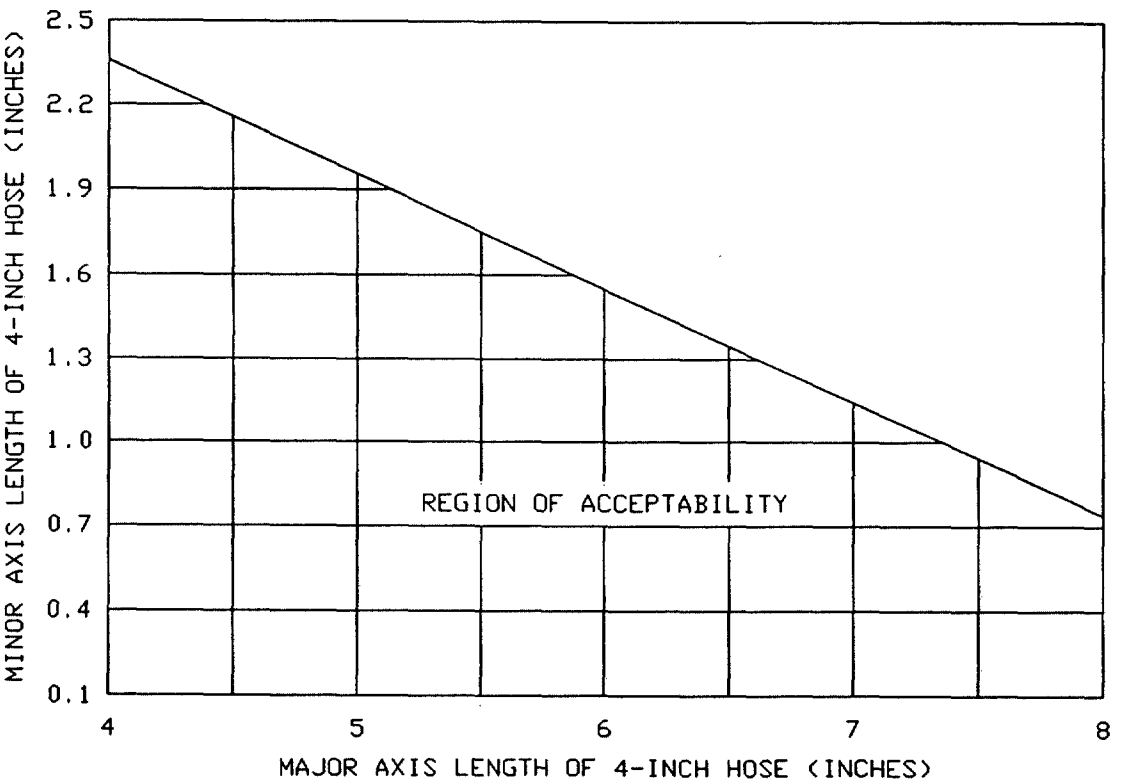



FIGURE 2. 4-inch hose collapsibility acceptance graph.



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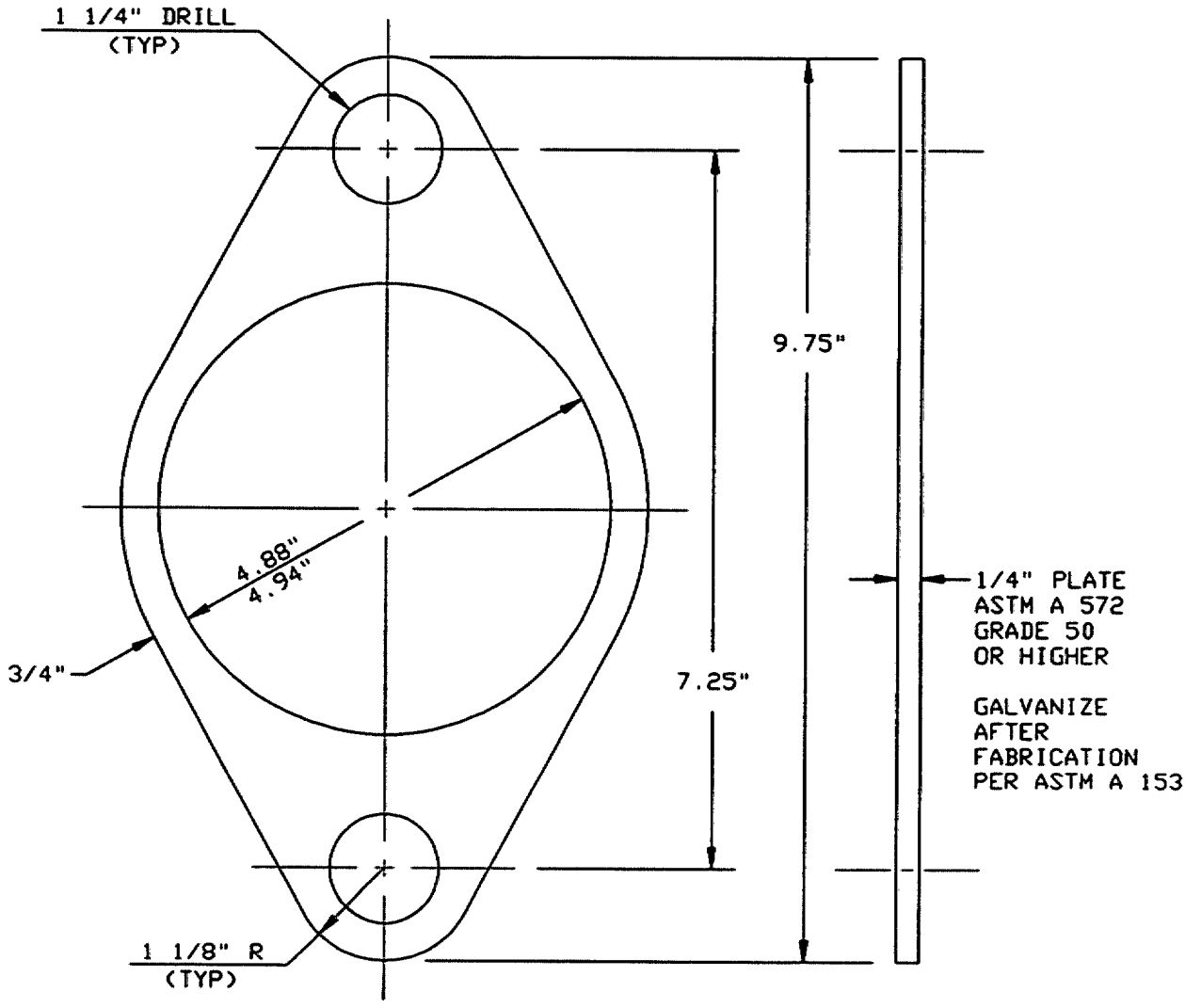



FIGURE 3. Ring, hose support Mark no. RHS - 4.



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# STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

## INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.
2. The submitter of this form must complete blocks 4, 5, 6, and 7.
3. The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

| I RECOMMEND A CHANGE: | 1. DOCUMENT NUMBER<br>MIL-H-28692(YD) | 2. DOCUMENT DATE *(YYMMDD)*<br>930406 |
|---|---|---|

3. DOCUMENT TITLE HOSE, NONMETALLIC, POTABLE WATER DISCHARGE, SMOOTH-BORE, LIGHTWEIGHT, 4 INCH, WITH REATTACHABLE COUPLINGS

4. NATURE OF CHANGE *(Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)*

5. REASON FOR RECOMMENDATION

| 6. SUBMITTER | | |
|---|---|---|
| a. NAME *(Last, First, Middle Initial)* | b. ORGANIZATION | |
| c. ADDRESS *(Include Zip Code)* | d. TELEPHONE *(Include Area Code)*<br>(1) Commercial<br>(2) AUTOVON *(If applicable)* | 7. DATE SUBMITTED *(YYMMDD)* |

| 8. PREPARING ACTIVITY | | |
|---|---|---|
| a. NAME<br>COMMANDING OFFICER | b. TELEPHONE *(Include Area Code)*<br>(1) Commercial<br>(805) 982-5782 | (2) AUTOVON<br>551-5761 |
| c. ADDRESS *(Include Zip Code)*<br>NCBC<br>CESO (Code 1564H) 621 Pleasant Valley Road<br>Port Hueneme, CA 93043-4300 | IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT:<br>Defense Quality and Standardization Office<br>5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466<br>Telephone (703) 756-2340 AUTOVON 289-2340 | |



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